QPCODE Reg. No:…….

Name: …….

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

THIRD SEMESTER MGU-UGP (HONOURS) B Sc (Hons) GEOLOGY EXAMINATION

OCTOBER 2025

MG3DSCGEO200 MINERALOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An)Evaluate(E),Create(C),Skill(S),Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer in 60 words

Answer any **7** questions

Each question carries **2** marks

1. What is the significance of streak of a mineral? [Ap][CO2]
2. Name two examples for oxide minerals [An][CO6]
3. What are miller indices? [U][CO1]
4. Differentiate polymorphism and isomorphism [Ap][CO2]
5. How is refractive index of a mineral useful in mineral identification? [Ap]CO3]
6. Write the role of Nicol Prism in optical mineralogy. [Ap][CO3]
7. What is pleochroism? Give an example. [Ap][CO3]
8. List the silicate groups of minerals [Ap][CO4]

[2x7=14]

Part B

Short notes in 250 words

Answer any 3 questions

Each question carries 8 marks

9. Write a brief note on physical and chemical properties of carbonate and sulphide mineral groups with examples [An][CO6]
10. Classification of minerals based on silicate structure [Ap][CO4]

11. Write a short essay on symmetry elements and forms present in the normal classes of Isometric system [U][CO1]
12. What are the major Physical properties of minerals and how they are useful in mineral identification? Explain with examples. [Ap] [CO2]
13. Write a note on feldspar group of minerals and how they are distinct from felspathoid group of minerals? [An][CO5]

[8x3=24]

Part C

Essay in 400 words

Answer any 1 question

Each question carries 12 marks

14. Analyze the properties of seven crystal systems and elaborate how their inherent symmetry controls physical and optical behaviour of minerals. Illustrate your discussion using specific mineral examples. [An][CO5]
15. Illustrate the working and components of a petrological microscope. Analyze the functions of optical accessories in observing and identifying minerals in thin sections. [Ap][CO3]

[12x1=12]

……..